PistonRH

ME Components RH Records

User Guide & Quick-Start Handbook

Running-hours tracking for piston, fuel valve, and exhaust valve components across your fleet - for vessel officers and technical office staff.

Prepared for internal circulation and demonstration

Contents

1. Who Uses This Tool

There are two kinds of accounts in PistonRH. Every user is set up by an administrator on purpose, so the right people see the right vessels.

Role	Can do
Vessel Officer	See and manage data only for the vessel(s) they've been given access to. Can also set up their own new vessel if they don't have one yet.
Technical Office	See and manage every vessel in the fleet. Only Technical Office can create user accounts and grant vessel access to Vessel Officers.

TIP: Not sure which one you are? Check the bottom-left corner of the sidebar once logged in - it shows your name and role.

2. Signing In for the First Time

1. Open the app's web address in your browser (given to you by your Technical Office).

2. You'll land on a public welcome page with a **Sign In** button. Click it.

3. Enter the email and password given to you by your administrator.

4. Click **Sign In**.

IMPORTANT: First time ever using the app, no account created yet? Someone with a Technical Office account needs to create your account first (see Section 14). There is no self-signup.

3. Setting Up Your Vessel

The first time you log in, if you don't yet have an active vessel, you'll see a **Setup First Vessel** screen instead of the Dashboard.

If you're Technical Office, you can either:

- Click **Load Demo Vessel** to explore the app with realistic sample data, or
- Fill in the **Create new vessel** form (vessel name, engine make, number of cylinders) and click **Create Vessel**.

If you're a Vessel Officer, you'll only see the **Create new vessel** form (the demo option is Technical Office-only). Fill it in and submit - you'll automatically get access to the vessel you just created.

Once a vessel exists and you have access to it, you'll land straight on the **Dashboard** every time you log in. If you manage more than one vessel, use the vessel switcher dropdown at the top of the sidebar to jump between them.

4. Key Concepts, in Plain Terms

"Unit"

Just marine shorthand for a cylinder number - Unit 1, Unit 2, etc. Each unit has one piston, one or more fuel valves, and one exhaust valve.

Overhaul Interval vs. Expected Life

The app uses two different labels depending on what kind of part you're looking at:

- **Overhaul Interval** - the threshold for a main component: a piston Unit, a Fuel Valve, or an Exhaust Valve. Shows **Overhaul Due** (or **Approaching**).
- **Expected Life** - the threshold for a sub-component nested under a main one (rings, nozzles/springs, seats/spindles). Shows **Life Exceeded** (or **Approaching**).

Parent / Child Components

A Fuel Valve or Exhaust Valve is the "parent." Its nozzle, spring, seat, or spindle can be linked to it as a "child." A child automatically inherits its parent's location, in-service status, and running-hours clock - you only track its own Expected Life threshold if it differs from the parent's part type. On any list, a child is shown indented underneath its parent with an "Inherited from ..." note.

Live RH

Running hours shown throughout the app are calculated automatically from your latest Monthly RH entry - no manual recalculation needed after logging a new reading.

5. The Dashboard

Your home screen after logging in. Three tabs: **Pistons**, **Fuel Valves**, **Exhaust Valves**.

- **Current ME RH** (top right) - the main engine's latest logged running hours reading.
- **Cylinder Matrix -** one card per Unit, showing its overall Overhaul Due/OK badge (based on the piston crown's overhaul cycle, consistent across all three tabs) and each fitted component with a progress bar toward its threshold.
- **Spares Overview** (fuel/exhaust tabs) - onboard spare components not currently fitted anywhere.

6. Logging Monthly RH

Go to **Monthly RH** in the sidebar. This is where you log the main engine's running-hour reading each month - the single most important regular task, since every other RH calculation in the app is derived from it.

1. Click **Add Entry**.
2. Enter the date and the ME's total running hours reading at that date.
3. Save.

> **TIP:** Every component's live RH across every page updates automatically the next time you view it - no manual recalculation needed.

7. Managing Components

Go to **Components**. Three tabs: Pistons, Fuel Valves, Exhaust Valves. This is your master inventory - add new parts, edit existing ones, or retire them.

Adding a piston component

1. Pistons tab -> **Add Component**.

2. Fill in Component ID, Type (Crown, Skirt, Rod, Ring...), Condition, Status, and location if it's currently fitted.

3. Save.

Adding a fuel valve or exhaust valve component

1. Fuel Valves (or Exhaust Valves) tab -> **Add Fuel Valve** / **Add Exhaust Valve**.

2. Fill in the same basic details.

3. If this is a nozzle, spring, seat, or spindle - not the valve body itself - use the **Parent Component** dropdown to pick which existing Fuel Valve/Exhaust Valve it belongs to. Leave it on "None - top-level component" for the valve body itself.

4. Save. A linked child appears indented under its parent, labelled "Inherited from [parent ID]."

Click the pencil icon on any row to edit it (including changing its Parent Component link later), or the trash icon to delete it permanently.

8. Recording Movements

Go to **Movements**. This is your audit trail - every time a component is fitted, removed, rotated between cylinders, landed ashore, or scrapped, record it here so the history and RH accounting stay accurate.

Action	Meaning
Fit	Moving a spare into service in a cylinder.
Remove	Taking a fitted component out to spare.
Rotate	Swapping a component from one cylinder to another.
Land Ashore	Sending it ashore (e.g. for reconditioning).
Receive Onboard	Bringing a part back onboard as a spare.
Scrap	Permanently retiring it.

Pick the Component, confirm From/To location, and the ME RH at the time of the movement (must be at or after your latest logged Monthly RH), then save. The full history is visible here and also in Reports -> History.

9. Cylinder Configuration

Go to **Cylinders**. This is where you manage which component is physically fitted in each cylinder slot, and set overhaul/dismantling baselines per cylinder.

- Each cylinder card shows what's currently fitted (Pistons tab), or the Fuel Valve/Exhaust Valve currently in each slot (Fuel/Exhaust tabs).
- **Fit Component to Cyl X** lets you register a brand-new component directly into that slot, or assign an existing spare already in your inventory.
- This page always shows the **parent** valve body in a slot **-** a nozzle or spring is never shown or assignable here directly (manage those from the Components page instead).
- **Baselines** (gear icon) lets you record the last overhaul and last dismantling running hours for that cylinder.

10. Alerts

Go to **Alerts** for a single combined list of every component across the vessel that's currently at Warning, Due/Approaching, or Overdue/Exceeded status - useful as a quick "what needs attention" check without digging through each tab.

11. Reports

Go to **Reports**. Four tabs:

- **Cylinder Status -** current running hours for every fitted piston, fuel valve, and exhaust valve component (with nested children), Unit by Unit.
- **Monthly RH -** the full history of logged ME readings.
- **Spares -** everything currently sitting onboard as a spare, by category.
- **History** - the full movement log (fit/remove/rotate/etc.) for pistons, fuel valves, and exhaust valves.

Use **Print View** for a clean printable layout, or **Export CSV** to download the data for use in Excel or elsewhere.

12. Settings

Go to **Settings**.

- **Vessel Profile -** name, IMO number, engine details, number of cylinders, and vessel-wide default Overhaul Interval/Warning thresholds for Units, Fuel Valves, and Exhaust Valves.
- **Component Type Thresholds -** override the vessel default for a specific component type. Types you've registered but haven't set a threshold for are flagged here automatically.
- **Maintenance Actions -** force a manual recalculation of every component's live RH.
- **Danger Zone -** reset all operational data or delete the vessel entirely. Both are irreversible.
- **User Management** (Technical Office only) **-** see Section 14.

13. Bulk Import From Excel

If you're setting up a vessel from scratch with a lot of existing components, it's much faster to import them from a spreadsheet than to type each one in by hand.

1. Go to **Import Setup**.

2. Click **Download sample template** to get a ready-made .xlsx file, pre-filled with example data for a full vessel - copy this format for your own data.

3. Open it and replace the example rows with your vessel's real data across the Vessel Info, Piston Components, Fuel Valves, and Exhaust Valves tabs.

4. To link a nozzle, spring, seat, or spindle to its parent valve, fill in the **Parent Component ID** column with the parent's Component ID (e.g. "FV1-1"), and leave that child row's own "Fitted In Cylinder" column blank.

5. Delete the yellow example rows before importing, don't change the column headers, and leave optional columns blank rather than typing "N/A."

6. Back in the app, drag your completed file into the upload area and click **Preview File**.

7. Review the preview - tick **Overwrite** on any conflicting Component ID you want to replace.

8. Click **Import** to confirm.

> **IMPORTANT:** Always grab a fresh copy of the sample template if it's been a while - an old saved copy on your computer won't have the latest columns if the app has been updated since.

14. Managing Users (Technical Office)

Only Technical Office accounts can do this. Go to **Settings** and scroll to **User Management**.

Creating a new account

1. Click **Add User**.
2. Fill in full name, email, a password (8+ characters), and pick the role - Vessel Officer or Technical Office.
3. Click **Create User**.

Giving a Vessel Officer access to a vessel

1. Find their row in the table - it shows how many vessels they have access to.
2. Click **Manage** next to their row.
3. Pick a vessel and click **+** to grant access, or the trash icon next to a listed vessel to revoke it.

Resetting a password or changing a role

1. Click the pencil icon on their row.
2. Update their name, role, and/or enter a new password (leave blank to keep it unchanged).
3. Click **Save Changes**.

> **TIP:** There's no self-service "forgot password." If someone's locked out, this is how you get them back in.

Flip the **Active** toggle off to temporarily block a login without deleting history, or use the trash icon to delete an account permanently. (You can't deactivate or delete your own account while logged in as it.)

15. Troubleshooting

"I forgot my password."

Contact your Technical Office administrator; they can reset it for you in Settings -> User Management.

"A form won't submit and nothing seems to happen."

Check for a small red message under one of the fields - this usually means a validation rule wasn't met (e.g. a password shorter than 8 characters).

"I just changed something but the page still shows the old data."

Try a hard refresh: Ctrl+Shift+R (Windows) or Cmd+Shift+R (Mac). The app caches data in your browser for speed.

"A component's status looks wrong."

Check Settings -> Component Type Thresholds and Vessel Profile first; a component's alert status depends on whichever threshold applies to it (its own override -> its type's override -> the vessel default).

"Imported nozzles/springs/seats/spindles aren't nesting under their parent valve."

The sheet didn't have the Parent Component ID column filled in for those rows (or an old template copy was used). Re-download the current sample template and re-import with Overwrite ticked for the affected rows.

Still stuck? Contact your Technical Office administrator or whoever manages this deployment for you.